

JAEHYUNG LEE

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EDUCATION

Johns Hopkins University, Baltimore, MD
Ph.D. in Materials Science Engineering

January 2025 – Present

Columbia University, New York, NY
M.S. in Chemical Engineering

Graduation: December 2024

Pennsylvania State University, State College, PA
B.S. in Chemical Engineering

Graduation: May 2023

WORK EXPERIENCE

EnerSys, Reading, PA

Jan 2022 – Jun 2022

Advanced Quality Engineer Intern

- Worked as an engineering intern performing failure mode effective analyses of Li-ion and lead-acid battery cells.
- Resolved a total of 204 production part deviation cases and gave monthly updates to project manager and C-suite.
- Partnered with diverse teams to generate a production line Li-ion battery control plan for a next-generation product.

United States Forces Korea, Camp Humphreys, South Korea

Jun 2018 – Feb 2020

KATUSA (E-5 Sergeant)

- Served 21 months of mandatory military service as a KATUSA (Korean Augmentation to the United States Army).
- Acted as a personal interpreter for a U.S. Army captain during joint Republic of Korea and U.S. Army operations.

RESEARCH EXPERIENCE

Johns Hopkins University, Baltimore, MD

Jan 2025 – Present

Doctoral Student

- Developing materials discovery workflows integrating DFT data with machine learning for energy applications.
- Building and applying neural network models to predict properties such as energies, forces, and electronic behavior.
- Designed agentic AI platforms that combine LLMs with scientific APIs to enable automated design workflows.
- Managing the group's Linux HPC server and optimizing DFT and ML workloads with GPU-accelerated inference.
- Curating and maintaining materials datasets for training, benchmarking, and validating machine learning models.

Columbia University, New York, NY

Jan 2024 – Dec 2024

Graduate Research Assistant

- Pursued atomistic scale research in Urban Lab, using computational tools to model and analyze materials behavior.
- Trained ML interatomic potentials of lithium nickel oxide and titanium oxide to perform efficient calculations.
- Awarded the Société de Chimie Industrielle Scholarship to conduct research and present at the poster session.

Pennsylvania State University, State College, PA

Aug 2020 – May 2023

Undergraduate Research Assistant

- Performed research of analyzing catalytic sites in metal oxides based on DFT results in Janik and Alexopoulos Labs.
- Mentored newly joined undergraduate students on how to utilize computational methods and machine learning tools.

TEACHING EXPERIENCE

Johns Hopkins University, Baltimore, MD

Jan 2025 – Present

- Teaching Assistant, EN.510.312 Physical Chemistry of Materials I: Thermodynamics
- Teaching Assistant, EN.520.677 Applied Quantum Computing

Spring 2026
Fall 2026

EXTRACURRICULAR ACTIVITIES

Columbia University Department of Chemical Engineering, New York, NY

Apr 2024 – Dec 2024

MS Ambassador

- Represented the department as an ambassador by communicating with admitted students and sharing program insight.
- Assisted the development of the MS handbook and participated as a panelist at the department master's colloquium.

Vice President

- Hosted monthly mentorship sessions and academic discussions among chemical engineering undergraduate students.
- Accompanied professors to represent Penn State University in the 2021 and 2022 AIChE National Conferences.

Engineers Without Borders PSU Chapter, State College, PA

Aug 2020 – May 2023

Development Committee Chair

- Led a subgroup in the club to design and facilitate the construction of a water supply facility in Namutamba, Uganda.
- Managed corporate networking and organized fundraising outreach events as chair of the development committee.

PUBLICATIONS

1. J. Lee, *et al.* **AGAPI-Agents: An Open-Access Agentic AI Platform for Accelerated Materials Design on AtomGPT.org.** *J. Phys. Chem. Lett.* **17**(26), 7221–7231 (2026). DOI: 10.1021/acs.jpcclett.6c00837.
2. J. Lee, *et al.* **Lessons Learned from the 2025 Agentic AI for Science Hackathon.** *Mach. Learn.: Sci. Technol.* **7**(4), 040501 (2026). DOI: 10.1088/2632-2153/ae7f6a.
3. J. Lee, *et al.* **BatteryMat: A hierarchical machine-learning and DFT framework for average-voltage screening of lithium-ion cathode materials.** *npj Comput. Mater.*, under review. Preprint: arXiv:2607.06645.
4. F. Abel, J. Lee, *et al.* **RamanGPT: Bidirectional Mapping Between Crystal Structures and Raman Spectra with Graph Neural Networks and Generative Transformers.** *Sci. Rep.*, under review. Preprint: arXiv:2606.03764.
5. A. Ajith, J. Lee, *et al.* **SlakoNet-VQD: A universal Slater-Koster tight-binding Hamiltonian for variational quantum band-structure calculations on near-term hardware.** *npj Quantum Inf.*, under review. Preprint: arXiv:2607.09872.
6. C. Campbell, J. Ely, J. Lee, *et al.* **Hybrid DiffractGPT-Rietveld Refinement Framework for Automated X-ray Diffraction Analysis.** *Sci. Rep.*, under review. Preprint: arXiv:2607.08890.
7. H. Neralla, J. Lee, *et al.* **Hallucination Detector: A hybrid LLM and Semantic Scholar tool calling for detecting hallucination in scientific literature on AtomGPT.org.** *Integr. Mater. Manuf. Innov.*, under review. Preprint: arXiv:2607.09774.
8. G. Han, T. Li, X. Xu, J. Lee, *et al.* **The search for high-entropy fuel-cell catalysts using disorder descriptors.** *Nano Futures* **9**(4), 045001 (2025). DOI: 10.1088/2399-1984/ae19b0.
9. B. Slautin, K. Barakati, U. Pratiush, C. Lowe, C. Bodinger, B. Cossairt, M. Ahmadi, A. Houton, T. Bazhurov, J. Lee, *et al.* **From Photons to Electrons: Accelerated Materials Discovery via Random Libraries and Automated Scanning Transmission Electron Microscopy.** *Microsc. Today*, under review. Preprint: arXiv:2603.20858.

SELECTED PRESENTATIONS

1. “SlakoNet DB: A Cross-Domain Tight-Binding Database of Electronic Structure,” Workshop on Recent Developments in Electronic Structure (ESW), Madison, WI, June 2026.
2. “Agentic AI for Materials Design with AGAPI,” Materials Research Society (MRS) Spring Meeting & Exhibit, Honolulu, HI, April 2026.
3. “BatteryMat: A Machine Learning Accelerated DFT Framework for Screening Battery Cathode Materials,” National Institute of Standards and Technology (NIST) Quantum Matters in Materials Science (QMMS) Workshop, Gaithersburg, MD, February 2026.
4. “Analysis of Cathode Materials for Lithium-Ion Batteries Using Machine Learning Potentials,” Société de Chimie Industrielle Biennial International Palladium Medal Ceremony, New York, NY, May 2024.